

```

106.
107.     elif move == 'P':
108.         # Let the player select a bucket to pour into:
109.         while True: # Keep asking until valid bucket entered.
110.             print('Select a bucket to pour into: 8, 5, or 3')
111.             dstBucket = input('> ').upper()
112.             if dstBucket in ('8', '5', '3'):
113.                 break # Player has selected a valid bucket.
114.
115.         # Figure out the amount to pour:
116.         dstBucketSize = int(dstBucket)
117.         emptySpaceInDstBucket = dstBucketSize - waterInBucket[dstBucket]
118.         waterInSrcBucket = waterInBucket[srcBucket]
119.         amountToPour = min(emptySpaceInDstBucket, waterInSrcBucket)
120.
121.         # Pour out water from this bucket:
122.         waterInBucket[srcBucket] -= amountToPour
123.
124.         # Put the poured out water into the other bucket:
125.         waterInBucket[dstBucket] += amountToPour
126.         steps += 1
127.
128.     elif move == 'C':
129.         pass # If the player selected Cancel, do nothing.

```

After entering the source code and running it a few times, try making experimental changes to it. On your own, you can also try to figure out how to do the following:

- Add variety by making the game configurable so you can specify any sizes for the three buckets and any amount for the goal quantity.
- Add a “hint” that examines the amount of water in each bucket and provides the next step to take. If the program can’t figure out which action to take next, it can simply display “I don’t know what you should do next. Maybe start over?”

Exploring the Program

Try to find the answers to the following questions. Experiment with some modifications to the code and rerun the program to see what effect the changes have.

1. What happens if you change `waterInBucket[srcBucket] = 0` on line 104 to `waterInBucket[srcBucket] = 1`?
2. What happens if you change `{'8': 0, '5': 0, '3': 0}` on line 16 to `{'8': 0, '5': 4, '3': 0}`?
3. What happens if you change `{'8': 0, '5': 0, '3': 0}` on line 16 to `{'8': 9, '5': 0, '3': 0}`?